PS Analysis in Data Science: Problem Set 2

Problem 1. Let X be a metric space and $f : [0,1] \to X$ continuous. Show that $f([0,1]) \subseteq X$ is path-connected.

Problem 2. Let $f : [0,1] \to \mathbb{R}^2$ and $f_1, f_2 : [0,1] \to \mathbb{R}$, so that

$$f(t) = \begin{pmatrix} f_1(t) \\ f_2(t) \end{pmatrix}$$

We equip $\mathbb{R}^2$ with the standard metric

$$d(x,y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}.$$

(a) Prove that the coordinate map $\chi_i : \mathbb{R}^2 \to \mathbb{R}$ defined by $\chi_i(x_1, x_2) = x_i$ is continuous, for $i = 1, 2$.

(b) Prove that f is continuous if and only if both f_1 and f_2 are continuous.

Problem 3. Let $S = \mathbb{R}^2 \setminus \{(0,0)\}$ and $x = (1,0)$ and $y = (-1,0)$.

(a) Define a continuous path from x to y.

(b) Plot this path in Julia.

(c) Visualize the sets

$$A = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}, \qquad B = \{(-1,y) : -1 \le y \le 1\} \cup \{(1,y) : -1 \le y \le 1\}.$$

in Julia by sample/grid points. Determine which of the two sets is path-connected.

Problem 4. Let $f : [0,1] \to \mathbb{R}$ be given by

$$f(x) = \frac{1}{4}(x^2 + 1).$$

(a) Show that f maps $[0,1]$ into itself and is a contraction on $[0,1]$.

(b) Deduce that f has a unique fixed point and implement the fixed point iteration in Julia.